Calculus BC Study Guide

Integration & Differential Equations

Seven focused lessons + seven 30-minute practice sets

How to use this packet

1. Study one topic at a time. Read the formal statement, then explain in your own words what it lets you do.
2. Cover the worked example and try it yourself. Compare each step, not just the final answer. This study time is **not** part of the 30 minutes.
3. Set a **30-minute timer** for that topic's two-page practice set. There are six problems per set; their suggested times total 30 minutes. Unless a set says otherwise, close the guide and work without a calculator.
4. If stuck, write a valid setup and move on. After the timer, use the separate worked-solutions packet. Correct the first incorrect step, then retry the problem without looking.

Practice time is **210 minutes total**, best split across seven sessions. Thirty minutes is a planning target; learning speed varies. Use extra paper if needed.

Topic map

6.1	Indefinite integral rules and initial conditions
6.2	LRAM, RRAM, MRAM; definite-integral properties; integral MVT
6.3	Inscribed/circumscribed rectangles and their infinite limits
6.4	Fundamental Theorem of Calculus (FTC)
6.6	Rectangle, trapezoidal, Simpson approximations; error estimates
7.1	Slope fields and Euler's method
7.2	Antiderivatives using u-substitution

Scope and references

The numbering above follows your class list, **not College Board's topic numbering**. Here, "infinite rectangles" means the limit of finite lower/upper rectangle sums, not an improper integral over an infinite interval.

AP alignment. College Board's *AP Calculus AB and BC Course and Exam Description*, Topics 6.2–6.9, 7.3–7.5, and 8.1, covers the related integration, slope-field, Euler, and average-value skills. Euler's method is BC-only. Simpson's rule and derivative-based quadrature error bounds are included as **classroom extensions**, not listed AP requirements. They are different from BC series error bounds.

Mathematical reference. OpenStax, *Calculus Volume 2*, Section 3.6, "Numerical Integration," for midpoint/trapezoidal/Simpson formulas and error bounds. Examples and practice problems here are original. Test tips are mathematical safeguards, not a promise of a particular scoring rubric.

6.1 — Indefinite Integral Rules

Formal statement / idea

An **antiderivative** of f on an interval is a function F satisfying $F'(x) = f(x)$ throughout that interval. If F is one antiderivative, then all are $F + C$:

$$\int f(x)\,dx = F(x) + C.$$

This is a *family of functions*, not a number or an area. Linearity gives $\int [af(x) + bg(x)]\,dx = a\int f(x)\,dx + b\int g(x)\,dx$ for constants a, b.

Integrand	**Antiderivative (add C)**
x^p, $p \neq -1$	$x^{p+1}/(p+1)$
$1/x$	$\ln\lvert x\rvert$ on an interval avoiding 0
e^x; a^x $(a > 0,\ a \neq 1)$	e^x; $a^x/\ln a$
$\cos x$; $\sin x$	$\sin x$; $-\cos x$
$\sec^2 x$; $\csc^2 x$	$\tan x$; $-\cot x$
$\sec x \tan x$; $\csc x \cot x$	$\sec x$; $-\csc x$
$1/(1+x^2)$; $1/\sqrt{1-x^2}$	$\arctan x$; $\arcsin x$ $(\lvert x\rvert < 1)$

Use each rule only where the integrand is defined. For a linear input, if $F' = f$ and $k \neq 0$, $\int f(kx+b)\,dx = F(kx+b)/k + C$.

Why it is used

Integration reverses differentiation: recover position from velocity, or a quantity from its rate. An initial condition selects one member of the family.

Classic worked example

Find F if $F'(x) = 6x^2 - 4/x + 3\cos x$, $x > 0$, and $F(1) = 5$.

$$F(x) = 2x^3 - 4\ln x + 3\sin x + C.$$

Insert the *function value*: $5 = 2 + 3\sin 1 + C$, so $C = 3 - 3\sin 1$. Differentiate the result to check: $6x^2 - 4/x + 3\cos x$ is recovered.

Test gotchas

- Rewrite roots/reciprocals as powers first. The power rule fails at $p = -1$.
- $\int \sin x\,dx = -\cos x + C$; the minus sign is easy to lose.
- Do not integrate a product by multiplying antiderivatives. Linearity applies to *sums*, not products or quotients.
- Indefinite integrals need $+C$; initial conditions determine it. Definite-integral values do not need $+C$.

6.1 — 30-Minute Practice

No calculator. Show your setup and reasoning. Use exact values unless directed otherwise. The time beside each problem is a pacing target, not a separate deadline.

1. Evaluate both integrals. *(4 min)*

$$\text{(a)} \int (8x^3 - 6x + 5)\,dx \qquad \text{(b)} \int \left(3\sqrt{x} - \frac{2}{x^2}\right) dx, \quad x > 0.$$

2. Evaluate both integrals on intervals where they are defined. *(5 min)*

$$\text{(a)} \int \left(\frac{4}{x} + 2e^x\right) dx \qquad \text{(b)} \int \left(3\sec^2 x - 2\sin x + \frac{1}{1+x^2}\right) dx.$$

3. Evaluate both integrals. *(4 min)*

$$\text{(a)} \int 6\cos(3x)\,dx \qquad \text{(b)} \int 5e^{-2x}\,dx.$$

6.1 — Practice, continued

4. Find $F(x)$ if $F'(x) = 3x^2 - 4x + 2$ and $F(1) = 5$. Verify your result by differentiation and substitution. *(5 min)*

5. A student writes

$$\int \left(\frac{1}{x} + \cos x\right) dx = \frac{\ln|x|}{x} + \sin x.$$

Differentiate the proposed answer, identify the errors, and give a correct general antiderivative. *(6 min)*

6. A particle has velocity $v(t) = 6t - 4$ meters per second and position $s(0) = 7$ meters. Find $s(t)$, its position at $t = 3$, and its displacement from $t = 0$ to $t = 3$. Explain why the last two answers differ. *(6 min)*

6.2 — Riemann Sums and Definite Integrals

Formal statement / idea

For a partition $a = x_0 < x_1 < \cdots < x_n = b$, write $\Delta x_i = x_i - x_{i-1}$ and choose $x_i^* \in [x_{i-1}, x_i]$. A Riemann sum is

$$\sum_{i=1}^{n} f(x_i^*)\Delta x_i.$$

If f is continuous on $[a, b]$, these sums approach the same number as the largest interval width tends to zero. That number is $\int_a^b f(x)\,dx$.

For n equal intervals, $\Delta x = (b-a)/n$ and $x_i = a + i\Delta x$:

$$L_n = \Delta x \sum_{i=1}^{n} f(x_{i-1}) \qquad \text{LRAM: left endpoint,}$$

$$R_n = \Delta x \sum_{i=1}^{n} f(x_i) \qquad \text{RRAM: right endpoint,}$$

$$M_n = \Delta x \sum_{i=1}^{n} f\left(\frac{x_{i-1} + x_i}{2}\right) \qquad \text{MRAM: midpoint.}$$

Plain language: width × height, added across every interval. For unequal widths, multiply *each* height by its own width.

Why it is used

A rate table may be all you have. Rectangle sums estimate the accumulated change even without an antiderivative. Rate units times input units give total-change units.

Classic worked example

For $f(x) = x^2 + 1$ on $[0, 2]$ with $n = 2$, $\Delta x = 1$:

	$[0, 1]$	$[1, 2]$	sum
Left heights	$f(0) = 1$	$f(1) = 2$	$L_2 = 3$
Right heights	$f(1) = 2$	$f(2) = 5$	$R_2 = 7$
Midpoint heights	$f(1/2) = 5/4$	$f(3/2) = 13/4$	$M_2 = 9/2$

Because f increases, $L_2 \le \int_0^2 f \le R_2$. The midpoint height is f(midpoint), **not** the average of endpoint heights.

Test gotchas

- n intervals need $n + 1$ endpoints but only n rectangle heights.
- Increasing: left under, right over. Decreasing: reverse. Without monotonicity, endpoint labels alone do not determine the error direction.
- A definite integral is *signed* accumulation. Geometric area uses $|f|$.
- A table of endpoints does not supply midpoint values unless those inputs are also listed. Do not invent them by averaging outputs.

6.2 — Properties, Average Value, and Integral MVT

Formal statements

For integrable f, g and constants α, β:

$$\int_a^a f\,dx = 0, \qquad \int_b^a f\,dx = -\int_a^b f\,dx,$$
$$\int_a^c f\,dx + \int_c^b f\,dx = \int_a^b f\,dx, \qquad \int_a^b (\alpha f + \beta g)\,dx = \alpha \int_a^b f\,dx + \beta \int_a^b g\,dx.$$

For $a < b$, if $f \le g$ then $\int_a^b f \le \int_a^b g$. In particular, if $m \le f(x) \le M$, then $m(b-a) \le \int_a^b f \le M(b-a)$. If f is even, $\int_{-a}^a f = 2\int_0^a f$; if f is odd, $\int_{-a}^a f = 0$ (assuming integrability).

Average value: for integrable f on $[a, b]$, $a < b$,

$$f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

Mean Value Theorem for Integrals: if f is *continuous* on $[a, b]$, $a < b$, there exists $c \in (a, b)$ such that

$$f(c) = f_{\text{avg}}, \qquad \text{equivalently} \qquad \int_a^b f(x)\,dx = f(c)(b-a).$$

Why it is used

Properties combine known integrals without re-integrating. Average value is the constant height giving the same signed area. Continuity guarantees that the function actually attains that average height somewhere in the interval.

Classic worked examples

If $\int_0^2 f = 5$ and $\int_2^5 f = -1$, then

$$\int_0^5 f = 4, \qquad \int_5^0 f = -4, \qquad \int_0^5 (2f+3)\,dx = 2(4) + 3(5) = 23.$$

For $f(x) = x^2$ on $[0, 3]$, $\int_0^3 x^2\,dx = 9$. Thus $f_{\text{avg}} = 9/3 = 3$. Since x^2 is continuous, the theorem applies. Solve $c^2 = 3$: only $c = \sqrt{3}$ lies in $(0, 3)$.

Test gotchas

- Average *function value* is $\int f/(b-a)$, not $(f(b) - f(a))/(b-a)$, which is average rate of change.
- The integral MVT uses $f(c)$, not $f'(c)$. State continuity when invoking it.
- Average the function over the interval, not merely its two endpoint values.
- $\int_a^b k\,dx = k(b-a)$: integrating a constant still introduces interval length.

6.2 — 30-Minute Practice

No calculator. Show your setup and reasoning. Use exact values unless directed otherwise. The time beside each problem is a pacing target, not a separate deadline.

1. For $f(x) = x^2$ on $[0, 2]$, compute L_2, R_2, and M_2. List the sample inputs used for each sum. *(6 min)*

2. Water enters a tank at a continuous, increasing rate $r(t)$ liters/minute.

t (min)	0	2	5	6
$r(t)$	1	3	4	5

   Using the three listed subintervals, estimate the amount entering with left and right sums. State which is a lower estimate, and include units. *(5 min)*

3. Suppose $\int_0^2 f\,dx = 3$, $\int_2^5 f\,dx = -4$, and $\int_0^5 g\,dx = 6$. Find (a) $\int_0^5 f\,dx$, (b) $\int_5^2 f\,dx$, and (c) $\int_0^5 (2f - g + 1)\,dx$. *(4 min)*

6.2 — Practice, continued

4. Find the average value of $f(x) = x^2 + 1$ on $[0, 3]$ and all $c \in (0, 3)$ satisfying the integral MVT conclusion. State why the theorem applies. You may use $\int_0^3 (x^2 + 1)\,dx = 12$. *(5 min)*

5. The graph of f forms a triangle above the x-axis on $[0, 2]$ with base 2 and height 3. On $[2, 4]$ it forms a semicircle below the axis with radius 1. Find (a) $\int_0^4 f\,dx$, (b) $\int_0^4 |f|\,dx$, and (c) the average value of f on $[0, 4]$. *(5 min)*

6. Let $f(x) = 0$ for $0 \le x < 1$ and $f(x) = 2$ for $1 \le x \le 2$. Find its average value on $[0, 2]$. Is there a $c \in (0, 2)$ where $f(c)$ equals that average? Does this contradict the integral MVT? Explain. *(5 min)*

6.3 — Inscribed and Circumscribed Rectangles

Formal statement / idea

Let f be continuous on $[a, b]$ and P a partition. On each subinterval set

$$m_i = \min_{[x_{i-1},x_i]} f, \quad M_i = \max_{[x_{i-1},x_i]} f, \quad \ell(P) = \sum_i m_i \Delta x_i, \quad U(P) = \sum_i M_i \Delta x_i.$$

Then $\ell(P) \le \int_a^b f\,dx \le U(P)$. If $f \ge 0$, these are the areas of **inscribed** and **circumscribed** rectangles. For signed f, call them lower/upper sums instead of ordinary rectangle areas.

As the largest subinterval width $\|P\|$ tends to 0,

$$\boxed{\lim_{\|P\|\to 0} \ell(P) = \int_a^b f(x)\,dx = \lim_{\|P\|\to 0} U(P).}$$

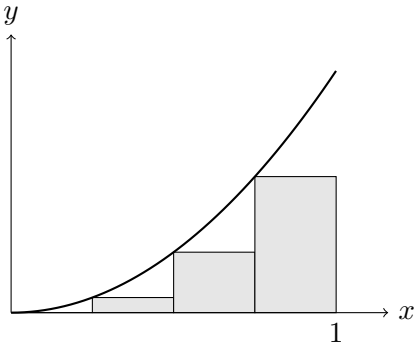
A continuous function is integrable, so the upper/lower gap tends to zero.

Why it is used

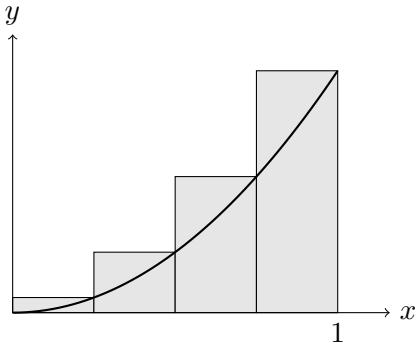
This makes "area under a curve" precise and traps the exact value between computable bounds. The infinite process is a *limit*; you never add literal rectangles of width zero.

Classic geometric example

For $f(x) = x^2$ on $[0, 1]$, increasing behavior means lower rectangles use left endpoints, and upper rectangles use right endpoints.



Lower: $\ell_4 = 7/32$



Upper: $U_4 = 15/32$

Test gotchas

- "Inscribed" does not always mean left. For decreasing $f \ge 0$, right endpoints give the inscribed rectangles.
- If a curve turns within an interval, its maximum/minimum may be inside, not at an endpoint.
- Merely saying $n \to \infty$ is not enough for arbitrary partitions: the *largest width* must tend to zero. Equal partitions satisfy this automatically.

6.3 — Turning a Rectangle Sum into an Exact Integral

Classic limit calculation

Continue with $f(x) = x^2$ on $[0, 1]$ and n equal intervals. Then $\Delta x = 1/n$, left inputs are $(i-1)/n$, and right inputs are i/n.

$$\ell_n = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{i-1}{n}\right)^2 = \frac{(n-1)n(2n-1)}{6n^3} = \frac{1}{3} - \frac{1}{2n} + \frac{1}{6n^2},$$
$$U_n = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{i}{n}\right)^2 = \frac{n(n+1)(2n+1)}{6n^3} = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}.$$

Both limits are $1/3$. Since $\ell_n \le \int_0^1 x^2\,dx \le U_n$, the squeeze theorem gives $\int_0^1 x^2\,dx = 1/3$. Notice $U_n - \ell_n = 1/n \to 0$.

Recognizing a limit of sums

For continuous f,

$$\int_a^b f(x)\,dx = \lim_{n\to\infty}\sum_{i=1}^{n} f\left(a + i\frac{b-a}{n}\right)\frac{b-a}{n}.$$

Read the expression in this order:

1. Identify the rectangle width $\Delta x = (b-a)/n$.
2. Identify the sample input $x_i = a + i\Delta x$.
3. Identify the function applied to that input.

For example,

$$\lim_{n\to\infty}\sum_{i=1}^{n}\sqrt{1+\left(2+\frac{3i}{n}\right)^2}\frac{3}{n} = \int_2^5 \sqrt{1+x^2}\,dx.$$

No evaluation is needed to translate the sum.

Useful finite-sum identities

$$\sum_{i=1}^{n} 1 = n, \qquad \sum_{i=1}^{n} i = \frac{n(n+1)}{2}, \qquad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}.$$

These simplify *finite* sums before taking the limit. Keep the width factor outside the sum all the way through.

Test gotchas

- The starting point a may be nonzero. A width of $3/n$ tells you the interval length is 3, not that the interval is $[0, 3]$.
- One rectangle sum is usually an approximation; the limit is the exact integral.
- When translating, keep the integrand and width together. A missing $1/n$ changes the problem completely.

6.3 — 30-Minute Practice

No calculator. You may use the finite-sum identities on the preceding page. Show your setup and reasoning. Use exact values unless directed otherwise. The time beside each problem is a pacing target, not a separate deadline.

1. For $f(x) = 4 - x$ on $[0, 4]$, use four equal subintervals. Draw the inscribed rectangles, compute their total area, and compute the circumscribed total. Identify which endpoint rule produces each. *(5 min)*

2. For $f(x) = x^2$ on $[0, 2]$, write ℓ_n and U_n as sigma sums with n equal intervals. Simplify both expressions in terms of n using the finite-sum identities. *(7 min)*

3. Take the limits of your two expressions from Problem 2 and state the exact area under x^2 on $[0, 2]$. Explain why the two limits determine the area. *(3 min)*

6.3 — Practice, continued

4. Write each limit as a definite integral; do not evaluate. *(5 min)*

$$\text{(a)}\ \lim_{n\to\infty}\sum_{i=1}^{n}\left[1+\left(2+\frac{3i}{n}\right)^2\right]\frac{3}{n} \qquad \text{(b)}\ \lim_{n\to\infty}\sum_{i=1}^{n}\cos\left(\frac{\pi i}{2n}\right)\frac{\pi}{2n}.$$

5. Write $\int_1^4 \sqrt{x+2}\,dx$ as a limit of right-endpoint sums using n equal subintervals. State Δx and x_i separately. *(4 min)*

6. Let $f(x) = 4-(x-2)^2$ on $[0,4]$, using the partition $0, 1, 3, 4$. Find the lower and upper sums. Is either the left sum or the right sum an upper sum for this partition? Justify by identifying the extrema on $[1,3]$. *(6 min)*

6.4 — Fundamental Theorem of Calculus

Formal statements

Accumulation form. If f is continuous on an interval containing a, then $A(x) = \int_a^x f(t)\,dt$ is differentiable at interior points and $A'(x) = f(x)$. With differentiable bounds and continuous f on the relevant interval,

$$\frac{d}{dx}\int_{u(x)}^{v(x)} f(t)\,dt = f(v(x))v'(x) - f(u(x))u'(x).$$

Evaluation form. If f is continuous on $[a, b]$ and $F' = f$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Why it is used

The FTC links accumulation to rate: differentiate accumulated change to recover the rate; integrate a rate to recover net change. It also replaces long limit-of-sums calculations with endpoint evaluation.

Classic worked examples

For $G(x) = \int_1^{x^2} \sqrt{1 + t^4}\,dt$, no elementary antiderivative is needed:

$$G'(x) = \sqrt{1 + (x^2)^4}\,(2x) = 2x\sqrt{1 + x^8}.$$

The outer FTC step supplies the integrand at the upper bound; the chain rule supplies $2x$.

For an evaluation,

$$\int_0^2 (3x^2 - 4x)\,dx = [x^3 - 2x^2]_0^2 = 0.$$

Zero is *net* signed area, not zero total area: this integrand changes sign.

If $H(x) = \int_0^x (t - 2)\,dt$, then $H' = x - 2$ and $H'' = 1$. Thus H decreases for $x < 2$, increases for $x > 2$, and has a local minimum at 2.

Test gotchas

- Differentiate an accumulation integral by substituting into the integrand, not by differentiating that integrand. $A' = f$, while $A'' = f'$ when f' exists.
- Moving upper bound: multiply by its derivative. Moving lower bound: also subtract that term.
- To find where A increases, inspect the *sign of* f, not whether f increases. For concavity of A, inspect f'.
- In $\int f(t)\,dt$, t is a dummy variable. The simple moving-bound formula assumes the integrand has no other dependence on x.
- A discontinuity inside an interval can invalidate direct use of the stated evaluation theorem. Do not blindly apply it across a vertical asymptote.

6.4 — 30-Minute Practice

No calculator. Show your setup and reasoning. Use exact values unless directed otherwise. The time beside each problem is a pacing target, not a separate deadline.

1. Evaluate both integrals. *(4 min)*

$$\text{(a) } \int_0^2 (3x^2 - 2x + 1)\,dx \qquad \text{(b) } \int_1^e \frac{1}{x}\,dx.$$

2. Let $A(x) = \int_2^x \sqrt{1 + t^3}\,dt$ for $x > 0$. Find $A'(x)$, $A'(2)$, and $A''(x)$. *(4 min)*

3. Differentiate both functions. *(5 min)*

$$\text{(a) } G(x) = \int_0^{x^3} \cos(t^2)\,dt \qquad \text{(b) } H(x) = \int_x^4 \ln(1 + t^2)\,dt.$$

6.4 — Practice, continued

4. Let $K(x) = \int_{x^2}^{3x} (1 + t^2)\, dt$. Find $K'(x)$ and $K'(1)$. *(5 min)*

5. Let $F(x) = \int_0^x (t^2 - 4)\, dt$ for $-3 \le x \le 3$. Find the open intervals where F increases and decreases, the locations of its interior local extrema, and the intervals where it is concave up and down. Justify with derivatives. *(7 min)*

6. An object's rate of change of mass is $m'(t) = 2t - 3$ grams/minute for $0 \le t \le 4$, and $m(0) = 10$ grams. Find its net mass change, its mass at $t = 4$, and the time when its mass is smallest on this interval. *(5 min)*

6.6 — Numerical Integration

Formal rules

Write $I = \int_a^b f(x)\,dx$, $h = (b-a)/n$, and $f_i = f(a+ih)$ for equal subintervals. Rectangle rules L_n, R_n, M_n are defined in 6.2. The composite trapezoidal rule is

$$T_n = \frac{h}{2}[f_0 + 2f_1 + \cdots + 2f_{n-1} + f_n] = \frac{L_n + R_n}{2}.$$

For **unequal widths**, use $T = \sum_{i=1}^{n} \frac{x_i - x_{i-1}}{2}[f(x_{i-1}) + f(x_i)]$.

Simpson's rule (classroom extension): equal widths and *even* n are required:

$$S_n = \frac{h}{3}\left[f_0 + 4\sum_{\substack{i=1\\ i\text{ odd}}}^{n-1} f_i + 2\sum_{\substack{i=2\\ i\text{ even}}}^{n-2} f_i + f_n\right].$$

The weights are $1, 4, 2, 4, 2, \ldots, 4, 1$; for $n = 2$ they are $1, 4, 1$. For continuous f, these composite rules converge to I as $h \to 0$ (even n for Simpson).

Why it is used

Numerical rules approximate integrals from data or when a usable antiderivative is unavailable. Rectangles use constant heights; trapezoids use straight-line tops; Simpson's rule uses quadratic pieces.

Classic worked example

Approximate $\int_0^2 x^3\,dx$ using $n = 2$, so $h = 1$, and values $0, 1, 8$:

$$T_2 = \frac{1}{2}[0 + 2(1) + 8] = 5, \qquad S_2 = \frac{1}{3}[0 + 4(1) + 8] = 4.$$

The exact integral is 4. Simpson is exact for polynomials of degree at most three; it is *not* always exact for other functions.

Error direction

Behavior throughout $[a, b]$	**Underestimate**	**Overestimate**
$f' \geq 0$	L_n	R_n
$f' \leq 0$	R_n	L_n
$f'' \geq 0$	M_n	T_n
$f'' \leq 0$	T_n	M_n

Equality can occur. Monotonicity controls left/right errors; concavity controls midpoint/trapezoidal errors. Concavity alone does not determine Simpson's error sign.

Test gotchas

Count intervals, not table entries. Do not use equal-width formulas on uneven data. Simpson needs an even number of intervals, not an even number of data values. Show the weighted sum before entering arithmetic into a calculator.

6.6 — Error Estimates and Guaranteed Accuracy

Formal bounds (classroom extension)

Let $a < b$, use equal widths, and write $E_Q = |I - Q_n|$ for **absolute error**. Assume the indicated derivatives are continuous on $[a, b]$, and that K_j is an upper bound for $|f^{(j)}(x)|$ on the *whole* interval.

Rule	Derivative needed	Guaranteed upper bound
L_n or R_n	$\lvert f'\rvert \le K_1$	$E_L, E_R \le \dfrac{K_1(b-a)^2}{2n}$
M_n	$\lvert f''\rvert \le K_2$	$E_M \le \dfrac{K_2(b-a)^3}{24n^2}$
T_n	$\lvert f''\rvert \le K_2$	$E_T \le \dfrac{K_2(b-a)^3}{12n^2}$
S_n (n even)	$\lvert f^{(4)}\rvert \le K_4$	$E_S \le \dfrac{K_4(b-a)^5}{180n^4}$

Why these are useful: the exact integral may be unknown, yet a derivative bound still guarantees accuracy. A bound $E_Q \le B$ places I inside $[Q_n - B, Q_n + B]$. It does *not* say the actual error equals B.

Classic worked example: choose the number of intervals

For $I = \int_0^2 x^2\,dx$, $f'' = 2$, so take $K_2 = 2$. To guarantee trapezoidal error at most 0.01,

$$\frac{2(2)^3}{12n^2} = \frac{4}{3n^2} \le \frac{1}{100} \quad \Longrightarrow \quad n^2 \ge \frac{400}{3}.$$

Since $11^2 < 400/3 < 12^2$, choose $n = 12$. With $n = 2$, the bound is $1/3$; indeed $T_2 = 3$ and $I = 8/3$, so the actual error is $1/3$ in this example.

For the rectangle bound, on one width-h interval, the derivative bound gives $|f(x) - f(x_{i-1})| \le K_1(x - x_{i-1})$. Integrate to get error at most $K_1h^2/2$; adding n such bounds yields $K_1(b-a)^2/(2n)$.

Test gotchas

- Bound the *absolute value* of the derivative everywhere, not just its value at a convenient endpoint. A larger valid bound is safe but less efficient.
- Solve the inequality, then round n *up*. For Simpson, round up again if necessary to make n even. For a strict tolerance $E < \varepsilon$, ensure the bound is strictly below it.
- Doubling n divides these bounds by 2, 4, 4, and 16, respectively; this is a statement about the bounds, not an exact prediction of every actual error.
- If $I \neq 0$, relative error is $|I - Q_n|/|I|$; percent error is 100 times that. Neither is the same as absolute error.

6.6 — 30-Minute Practice

Calculator allowed for arithmetic only. You may use the rule and error-bound formulas. Show your setup and reasoning. Use exact values unless directed otherwise. The time beside each problem is a pacing target, not a separate deadline.

1. For $\int_0^2 x^2\,dx$, compute L_2, R_2, M_2, T_2, S_2. Compare each with the exact value $8/3$ and state whether it is low, high, or exact. *(7 min)*

2. A rate $r(t)$ in liters/minute is recorded below. Use trapezoids on the listed intervals to approximate $\int_0^6 r(t)\,dt$, including units. *(3 min)*

t	0	1	3	6
$r(t)$	2	4	3	5

3. A twice-differentiable function satisfies $f'(x) < 0$ and $f''(x) > 0$ throughout $[a, b]$. Classify L_n, R_n, M_n, and T_n as underestimates or overestimates. Explain which condition controls each pair. *(3 min)*

6.6 — Practice, continued

4. For $f(x) = x^4$ on $[0, 1]$, find valid bounds K_1, K_2, K_4. Use $n = 4$ to find absolute-error bounds for left/right rectangles, midpoint, trapezoidal, and Simpson's rule. *(6 min)*

5. For a smooth function on $[0, 2]$, suppose $|f''| \leq 3$ and $|f^{(4)}| \leq 10$. Find the smallest allowed n certified by the respective formulas to guarantee error at most 0.001 for (a) the trapezoidal rule and (b) Simpson's rule. *(6 min)*

6. A numerical estimate is $Q = 2.4$ with a guaranteed absolute-error bound 0.03. (a) Give an interval guaranteed to contain the exact integral. (b) A later exact calculation gives $I = 2.38$. Find the actual absolute error and percent error (round percent to three decimal places). (c) A student says $|I - Q| = 0.03$ must hold. Explain the mistake. *(5 min)*

7.1 — Slope Fields

Formal statement / idea

For a first-order differential equation

$$\frac{dy}{dx} = F(x, y),$$

a **slope field** places a short line segment of slope $F(x_0, y_0)$ at each selected point (x_0, y_0). A solution curve satisfies $y'(x) = F(x, y(x))$ at every point and is tangent to those local directions. An initial condition specifies a point the desired solution must pass through.

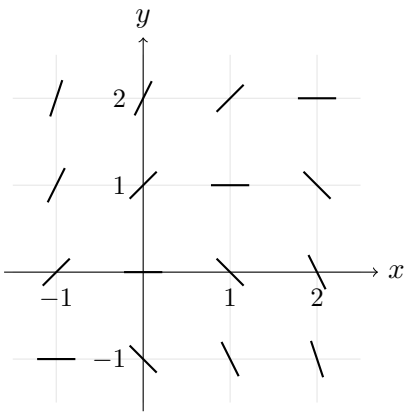
Equilibrium: $y(x) = k$ is a constant solution if $F(x, k) = 0$ for *every* x under consideration. A single horizontal segment does not establish an equilibrium.

Why it is used

A field shows solution behavior without solving the equation: where solutions rise or fall, where they level off, and how an initial value selects a path.

Classic worked example

For $y' = y - x$, the slopes at $(0, 1), (1, 1), (1, 0)$ are $1, 0, -1$. Every point on $y = x$ has slope zero, but $y = x$ is *not* a solution: that curve's derivative is 1, not 0.



Test gotchas

- Substitute *both* coordinates. The output is a slope, not a new y-coordinate.
- If F depends only on x, slopes repeat vertically. If only on y, they repeat horizontally.
- A solution must follow the field smoothly, not connect the centers of all its segments.
- For uniqueness, a standard sufficient hypothesis is that F and $\partial F/\partial y$ are continuous near the initial point. Under it, solution curves cannot cross there. Do not claim noncrossing without appropriate hypotheses.

7.1 — Euler's Method

Formal rule

Given $y' = F(x, y)$ and $y(x_0) = y_0$, choose step size h and set

$$\boxed{x_{k+1} = x_k + h, \qquad y_{k+1} = y_k + hF(x_k, y_k).}$$

This is a numerical approximation, based on the tangent-line model $y(x + h) \approx y(x) + h\,y'(x)$.

Why it is used

Euler follows the slope field one short tangent step at a time to estimate a solution value without finding an exact solution formula.

Classic worked example

For $y' = x + y$, $y(0) = 1$, estimate $y(0.5)$ using two equal steps. Here $h = (0.5 - 0)/2 = 0.25$.

k	x_k	y_k	$F(x_k, y_k)$	$y_{k+1} = y_k + hF(x_k, y_k)$
0	0	1	1	1.25
1	0.25	1.25	1.5	1.625

Thus $y(0.5) \approx 1.625$. The second slope uses the *updated* point $(0.25, 1.25)$, not the initial y again.

Concavity and error

A tangent line lies below a concave-up solution and above a concave-down solution. This determines the error direction of a *single tangent step from an exact point.* For multiple Euler steps, justify propagation of the error for the given equation; a slope field sketch alone is not a universal error proof.

For $y' = y$, $y(0) = 1$, forward Euler uses $y_{k+1} = (1+h)y_k$. The exact solution is e^x. For $h > 0$, $1+h < e^h$, so $(1 + h)^k < e^{kh}$: this particular forward-Euler sequence underestimates the solution.

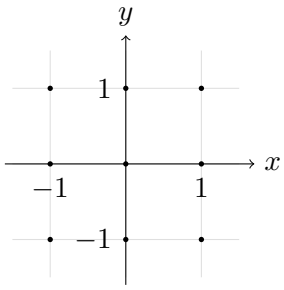
Test gotchas

- New value = old value + step size × current slope. A slope by itself is not the next value.
- Recompute the slope at every new (x_k, y_k). Show a table so the updates are visible.
- To move left, h is negative. Count updates, not rows: two steps require three solution points.
- Keep unrounded values during the iteration. Round only the requested final approximation.
- If $y' = F(x, y)$, then along a solution $y'' = F_x + F_y\,y'$. Substitute the differential equation when studying concavity.

7.1 — 30-Minute Practice

Calculator allowed for arithmetic only. Use exact fractions or terminating decimals; do not round intermediate values. Show your setup and reasoning. Use exact values unless directed otherwise. The time beside each problem is a pacing target, not a separate deadline.

1. For $y' = x + y$, draw a slope field at all nine points with $x, y \in \{-1, 0, 1\}$. Then sketch a short solution curve through $(0, 0)$. State its slope at $(0, 0)$ and whether that solution is constant. *(6 min)*



2. A field has equal slopes along each horizontal row. Its segments are horizontal at $y = 2$, positive below $y = 2$, and negative above $y = 2$. Which equation matches: $y' = x$, $y' = 2 - y$, or $y' = y - 2$? Justify, and identify an equilibrium solution. *(3 min)*

3. Given $y' = x + y$, $y(0) = 1$, use two Euler steps of size 0.5 to estimate $y(1)$. Show the current point, slope, and update for each step. *(6 min)*

7.1 — Practice, continued

4. Given $y' = x - y$, $y(1) = 2$, use two equal Euler steps to estimate $y(0)$. Show the signed step size and both updates. *(5 min)*

5. Given $y' = 2 - y$, $y(0) = 0$: (a) use three Euler steps of size 0.5 to estimate $y(1.5)$; (b) derive y'' in terms of y and determine the solution's concavity while $y < 2$. *(5 min)*

6. Given $y' = y$, $y(0) = 1$: (a) estimate $y(1)$ with one Euler step of size 1; (b) repeat with two steps of size 0.5; (c) given the exact solution $y = e^x$, compare both estimates with $e \approx 2.71828$ and decide which is closer. *(5 min)*

7.2 — Antiderivatives by u-Substitution

Formal statement / idea

Suppose g is continuously differentiable and f is continuous on an interval containing the values of g. If $F' = f$, the reverse chain rule gives

$$\int f(g(x))g'(x)\,dx = F(g(x)) + C.$$

Write $u = g(x)$ and $du = g'(x)\,dx$ to expose the basic integral. For definite integrals,

$$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du.$$

For this substitution identity, g need not be increasing; the transformed limits carry the direction.

Why it is used

A complicated expression may really be a basic antiderivative with an inside function. Substitution works when its derivative appears, possibly multiplied by a constant.

Classic worked examples

For $\int x(x^2+4)^3\,dx$, choose $u = x^2 + 4$. Then $du = 2x\,dx$, so $x\,dx = du/2$:

$$\int x(x^2+4)^3\,dx = \frac{1}{2}\int u^3\,du = \frac{u^4}{8} + C = \frac{(x^2+4)^4}{8} + C.$$

Check: differentiating yields $(1/8)\cdot 4(x^2+4)^3 \cdot 2x = x(x^2+4)^3$.

For $\int_0^1 \frac{2x}{1+x^2}\,dx$, use $u = 1 + x^2$, $du = 2x\,dx$. The limits become $u(0) = 1$, $u(1) = 2$, so

$$\int_1^2 \frac{1}{u}\,du = [\ln u]_1^2 = \ln 2.$$

This is the pattern $\int g'(x)/g(x)\,dx = \ln|g(x)| + C$ on an interval where $g \neq 0$.

Test gotchas

- Write *du* before integrating. Supply a missing *constant* factor correctly; do not invent a missing variable factor.
- In a rewritten u-integral, every x must be eliminated. An x left over signals unfinished substitution.
- For an indefinite integral, return to x and add C. For a definite integral, either change the limits to u or return to x before using the original limits. Never mix them.
- $\int \cos(x^2)\,dx$ is not handled by simply writing $\sin(x^2)/(2x)$: the derivative factor $2x$ is absent. Not every integral is a direct substitution problem.

7.2 — 30-Minute Practice

No calculator. Show the substitution and differential in each applicable problem. Show your setup and reasoning. Use exact values unless directed otherwise. The time beside each problem is a pacing target, not a separate deadline.

1. Evaluate both integrals. *(5 min)*

$$\text{(a)} \int 2x(x^2+5)^4\,dx \qquad \text{(b)} \int 3\cos(3x+1)\,dx.$$

2. Evaluate both integrals. *(5 min)*

$$\text{(a)} \int \frac{x}{x^2+9}\,dx \qquad \text{(b)} \int \frac{e^x}{1+e^x}\,dx.$$

3. Evaluate both integrals. *(5 min)*

$$\text{(a)} \int \frac{\sin x}{\sqrt{2+\cos x}}\,dx \qquad \text{(b)} \int \frac{6x}{1+9x^4}\,dx.$$

7.2 — Practice, continued

4. Evaluate $\int_0^2 x\sqrt{x^2+5}\,dx$ using transformed limits. *(5 min)*

5. Find $F(x)$ if $F'(x) = 2xe^{x^2}$ and $F(0) = 3$. Verify both the derivative and the initial condition. *(4 min)*

6. A student claims $\int \cos(x^2)\,dx = \sin(x^2)/(2x) + C$ for $x \neq 0$. (a) Differentiate the proposed antiderivative to test the claim. (b) Explain why $u = x^2$ does not turn this integrand directly into $\int \cos u\,du$. (c) Evaluate the related integral $\int x\cos(x^2)\,dx$. *(6 min)*